FRONTEND: THEORY ASSIGNMENT 2

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1. What is CSS Box Model?

Ans: The CSS Box Model explains how the size of an HTML element is calculated. It has four parts: Content, Padding, Border and Margin. Content is the actual text or image, padding is the space around the content, border surrounds the padding, and margin is the outer space between elements.

2. What is media query? Explain with a sample code.

Ans: A media query is used to apply different CSS styles for different screen sizes or devices. It helps to make a webpage responsive. For example:

```
@media (max-width: 600px) {
    body {
        background: lightblue;
        font-size: 14px;
    }
}
```

Here the style inside the media query is applied when the screen width is 600px or less.

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3. What are the features of Bootstrap?

Ans: Bootstrap is a popular front-end CSS framework. Its main features are responsive design, mobile-first approach, ready-made components, grid system, utility classes, buttons, forms, cards and navigation bars. It also provides support for different screen sizes.

4. Write an example of basic grid structure in Bootstrap.

Ans: Bootstrap divides a row into 12 columns. A basic example is:

```
<div class="container">
  <div class="row">
    <div class="col-md-6">Column 1</div>
    <div class="col-md-6">Column 2</div>
  </div>
</div>
```

Here the row contains two columns. Each col-md-6 takes 6 out of 12 columns, so both columns together use the full row.

5. What is CSS Selectors?

Ans: CSS selectors are used to select HTML elements on which CSS styles are applied. Common selectors are element selector, class selector, id selector, universal selector and group selector. Example: p selects all paragraph elements, .box selects elements with class box, and #header selects the element with id header.

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6. What are CSS backgrounds, list the properties?

Ans: CSS background properties are used to control the background of an element. Important properties are background-color, background-image, background-repeat, background-position, background-size, background-attachment and the shorthand background property.

Example:

```
body {
    background-color: lightblue;
    background-image: url("image.jpg");
    background-repeat: no-repeat;
    background-size: cover;
}
```

7. How can we overlap elements in CSS?

Ans: Elements can be overlapped by using position and z-index. Usually position: absolute or relative is used to place an element over another element. The z-index property controls which element appears in front.

Example:

```
.box1 {
    position: relative;
    z-index: 1;
}
.box2 {
    position: absolute;
    z-index: 2;
}
```

Since box2 has a higher z-index, it can appear above box1 when their positions overlap.

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8. What are the different positioning properties in CSS?

Ans: The main position values in CSS are static, relative, absolute, fixed and sticky. Static is the default position. Relative moves an element relative to its normal position. Absolute positions an element relative to its nearest positioned ancestor. Fixed keeps an element fixed with respect to the viewport. Sticky behaves like relative until a given scroll position is reached, then it sticks.

9. What is ECMAScript?

Ans: ECMAScript is a standard that defines the rules and features of scripting languages such as JavaScript. JavaScript follows the ECMAScript standard. It defines features such as variables, functions, objects, classes, arrays and many other language features. New versions add improvements to the language.

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10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone.

Ans: var, let and const are used to declare variables. var is function-scoped and can be redeclared. let and const are block-scoped and cannot be redeclared in the same block. let can be reassigned, while const cannot be reassigned after initialization.

All three are hoisted, but let and const stay in the Temporal Dead Zone (TDZ) from the start of the block until the declaration is executed. Accessing them before declaration causes a ReferenceError.

Difference:

Scope	: var = function	let = block	const = block
Redeclare	: var = Yes	let = No	const = No
Reassign	: var = Yes	let = Yes	const = No
TDZ	: var = No	let = Yes	const = Yes

Simple TDZ example:

```
console.log(x);   // ReferenceError
let x = 10;
```

The error happens because x is in the Temporal Dead Zone before the let declaration is reached.

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